



Post-Quantum Mathematics

State of Affairs Since the August Edition

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

This report covers post-quantum mathematics from the last full edition, in early September, to 4 October 2026. It follows the layer underneath the migration: where the new algorithms come from, who is choosing them, and what is being broken. Four weeks is a short window. It was a busy one for attacks, and for the bodies that answer them.

The report opens with the few developments that change the picture. It then turns to the code-based schemes, where Germany has changed its advice on Classic McEliece. The multivariate signature candidates come next, followed by NIST's own standards and its second signature competition. National tracks in China, South Korea and Europe follow. Isogenies, lattice variants and symmetric research close the new material.

The final part steps back. It describes where things stood before September, for anyone who has not read the earlier editions. Sources are numbered in the text and listed at the end.

1. What Matters Most

Most of what happened in four weeks refreshes the picture, and this section names the few things that change it.

Five things changed the picture this month. Germany stepped back from a code-based scheme it had recommended since 2020. South Korea's chosen signature scheme fell to a forgery, and a new attack technique swept across the multivariate signature candidates. China's new competition met the public and lost several designs within days. And NIST settled how it will specify the signature standard that has taken longest to finish.



Start with Germany. Its federal security office, BSI, now advises against choosing Classic McEliece for new developments or planning [1]. The advice sits in a separate note dated 29 September, announced in a press release on 1 October [1] [2]. It cites recent improvements in structural cryptanalysis, while noting that no practical break of the recommended parameters exists yet [1]. BSI will revise its main technical guideline to match, with the next version planned for early 2027 [2].

The shift reverses a long-standing position. Germany's guideline had recommended Classic McEliece since 2020, before NIST made its own selections [3]. NIST never chose the scheme, but ISO standardized it anyway [4].

Korea's own process took a direct hit. AIMER v2.0 is the signature scheme selected in South Korea's national KpqC competition [5]. A 27 September preprint gives a classical forgery attack on it that needs only the public key [5]. In the paper's own benchmark, the median time to forge an AIMER-256f signature is about 11.6 milliseconds [5]. A selected national scheme broken outright is a rarer event than a candidate losing bits.

From policy to mathematics. A new attack built on Schur-Macaulay matrices applies to every round-2 parameter set of UOV, MAYO, SNOVA and QR-UOV but one [6]. Four alternate MAYO sets and seven of eleven SNOVA sets fall below their claimed security [6]. After a separate structural forgery on MAYO, its round-2 Level I set sits 15 bits short of its target [7]. Even the round-3 set MAYO2 now falls 3 bits short [7].

Then China. Its Institute of Commercial Cryptography Standards published 119 first-round candidates for its new NGCC competition on 20 September [8]. Within three days, 104 public findings had been reported against 65 candidates, including breaks in five designs [8]. A preprint counting the first seven days found 191 findings against 89 candidates, 78 of them rated Critical [9]. An open competition that loses designs in its first week is doing its job early.

Finally, NIST. On 28 September it updated its plan for FIPS 206, the standard for the FN-DSA signature scheme [10]. The signing procedure will be specified in a single version, using fixed-point arithmetic only, in enough detail to validate against known-answer tests [10]. The plan follows recent fixed-point implementations of Falcon, the scheme's earlier name [10]. Key generation may be validated another way [10].

2. Code-Based Schemes: Classic McEliece and HQC

Code-based schemes rest on the difficulty of decoding a random-looking error-correcting code, and they are the main alternative to the lattice mathematics behind ML-KEM.

The attack that moved Germany arrived early in the month. An 11 September preprint turns an earlier distinguisher, known as GIJS, into full key-recovery attacks on Classic McEliece [11]. A distinguisher only tells an attacker that a key is a McEliece key rather than random, so key recovery is a large step beyond it. Under heuristic assumptions, the new attacks cost between 2^{94} and 2^{102} bit operations [11]. The best decoding attacks had cost 2^{151} or more, and the authors demonstrate concrete recovery on a challenge instance set by TII [11].

BSI's note spells out what remains allowed. Its current guideline recommends Classic McEliece only in hybrid use, combined with a key negotiated by a classical scheme [1]. For key agreement it now points to FrodoKEM or ML-KEM, and to HQC once that standard is published [1]. The note also records that ISO/IEC standardized

Classic McEliece in June 2026 [1]. So within four months the scheme gained an international standard and lost Germany's backing for new work.

HQC, NIST's chosen code-based backup, met a computational record of its own. On 30 September a record computation solved a weight-64 instance of HQC's quasi-cyclic decoding problem in about an hour on 24 GH200 GPUs [12]. Updated HQC security estimates are said to be forthcoming [12]. The standard itself has still not appeared. As of August, NIST had published no draft, despite earlier expecting one around March 2026 [13].

A smaller code-based design fell outright. A preprint recovers the full secret key of a knapsack-based McEliece variant in seconds, for all 24 proposed parameter sets, using the LLL lattice-reduction algorithm [14]. The family's long record belongs to Classic McEliece itself, not to every scheme built in its style.

Unresolved Conflicts

Two readings of the McEliece evidence sit side by side. The new key-recovery paper lowers estimated costs far below earlier attacks and demonstrates recovery on a challenge instance [11]. BSI says no practical attack yet exists on the recommended parameter sets, and still allows hybrid use [1]. Neither source says how close the challenge instance is to those sets. A demonstrated attack on a recommended set, or a fresh cost estimate for each one, would settle how urgent the shift is.

3. Multivariate Signatures Under Pressure

Multivariate schemes build signatures from systems of quadratic equations, and four of the candidates left in NIST's signature round belong to this family.

This month brought the heaviest pressure yet on the family. The Schur-Macaulay attack covers almost the whole round-2 field, and some SNOVA sets lose further ground through additional complexity reductions [6]. MAYO responded quickly. It discarded one round-3 parameter set and raised the others [6].

MAYO also took the most specific damage. The geometric-forgery paper draws on two earlier techniques, known as whipping and the Pseudo-Oil Attack [7]. It cuts round-2 Level I and Level III security by about 18 and 16 bits [7]. Round-3 sets lose between 2 and 6 bits each [7].

The attacks also reached the family's variants. One preprint extends divided-power attacks on UOV to arbitrary characteristic, aimed at QR-UOV [15]. Another forges signatures for Frobenius-UOV by solving a single univariate equation efficiently [16]. It needs roughly 2^{45} to 2^{53} field operations against instances meant to provide 128 to 256 bits of security [16].

SNOVA faced a second, separate attack. On 22 September two researchers reported a key-recovery attack on SNOVA parameter sets over fields of odd characteristic, posted to NIST's public forum [17]. It exploits vinegar variables, random values chosen during signing, that are not sampled uniformly [17]. Their proof of concept recovers the secret key for the SL1 and SL3 sets in between 22 seconds and 16 minutes [17]. It needs between 9 million and 180 million signatures to do so [17].

The SNOVA team answered on 28 September. It said the attack affects no recommended round-3 parameter set, because those are defined over fields of characteristic 2 [17]. It published an updated specification, version 3.1, on GitHub, and said its round-3 submission to NIST is unchanged [17].

Unresolved Conflicts

The two sides describe the same parameter sets in different terms. The attackers called them round-3 SNOVA parameter sets over fields of odd characteristic, while the SNOVA team calls them alternative sets, not recommended ones [17]. Both statements stand in the same forum thread. A statement from NIST on which SNOVA sets it is evaluating in round 3 would settle it.

4. NIST: The Signature Round and the Standards

NIST's process decides which algorithms most of the world will deploy, and this month moved both its second signature competition and its published standards.

The signature round now has eight candidates. NIST's status report IR 8610, from May, advanced nine: FAEST, HAWK, MAYO, MQOM, QR-UOV, SDiH, SNOVA, SQIsign and UOV [18]. HAWK was withdrawn on 29 July after a mathematical vulnerability was found [19].

Two of the remaining candidates drew a formal comment on 4 October. It argues that the security proofs of SDiH and MQOM assume the ideal cipher model [20]. That model treats a block cipher as an idealized random one. The comment calls the assumption questionable here, because AES-192 and AES-256 are vulnerable to related-key attacks [20]. It recommends more conservative assumptions in the proofs or the design, and says the advice is precautionary rather than a response to any attack [20].

SQIsign, the round's only isogeny scheme, got a sizing study. A 26 September preprint evaluates a dimension-4 variant at round-3 parameters [21]. It reports 142-byte signatures and 84-byte public keys at NIST security level 1, 226 bytes combined [21]. The price is a higher verification cost than the standard round-3 submission [21].

Lattice signatures had a mixed month. A 29 September preprint presents full key-recovery attacks on the original CRYSTALS-Dilithium and its round-1 and round-2 NIST versions [22]. The attacks work in the superposition model, where an attacker can query the scheme in quantum superposition. The same paper says changes made during the NIST process block them against the final ML-DSA standard [22]. Separately, a preprint called FALCON++ estimates signatures of 419 and 921 bytes, about 37 and 28 percent smaller than standard Falcon [23].

The standards around the signatures moved too. NIST's draft SP 800-230, from April, adds six SLH-DSA parameter sets for uses that need only a limited number of signatures [24]. Comments on it were made public by 1 July [24]. That corrects the August edition, which found nothing queued for the hash-based standard [25]. At the IETF, the composite ML-DSA signatures draft, approved by the IESG in April, entered its second RFC Editor edit on 2 October [26].

5. National Tracks: China, South Korea and Europe

Other countries run their own selection processes, and their requirements keep diverging even where the algorithms converge.

China's competition produced the month's numbers. The 119 first-round candidates comprise 34 signatures, 41 key encapsulation mechanisms, 9 key exchange schemes and 35 hashes [8]. The seven-day count comes from a preprint whose author says an AI-assisted evaluation effort found, checked and disclosed 110 of the 191 issues [9]. Forty-seven of those were rated Critical [9]. Beside the competition, China's industry ministry formed a quantum information standards committee, which first met on 30 July with 62 members [27].

The two tallies cover different windows, three days and seven, so the rising numbers reflect continued work rather than a disagreement [8] [9]. A finding is not the same as a break. Only five designs were reported broken in the first three days [8].

Several of those breaks now have papers behind them. One shows that BAG-Loong's reference implementation samples its secret matrices from fixed public subspaces, so Gaussian elimination recovers the key [28]. All 40 of its known-answer-test records, across four parameter sets, give up the exact secret [28]. Another recovers Polar-KEM's secret orthogonal matrix from the public key with a single linear-algebra operation, in polynomial time [29].

A third paper recomputes how often the lattice key encapsulation candidates fail to decrypt, which matters because decryption failures can leak secrets [30]. For Cheetah128 and Cheetah256 it estimates failure rates of $2^{-79.1}$ and $2^{-51.8}$, against claimed rates of 2^{-129} and lower [30]. The cause is a compression-noise term that the specification omits [30]. Correlated decoding blocks also raise the estimated failure rate of six of DTRU's seven parameter sets by roughly 50 to 110 bits [30].

South Korea set dates. On 17 September its National Intelligence Service announced a plan that brings post-quantum cryptography into public cybersecurity assessments from 2027 [31]. For key encapsulation, verification targets come in January 2027, verification in January 2028, and mandatory implementation by January 2029 [31]. Signatures follow a phased schedule, starting with verification targets in January 2028 [31].

Korea is also building public engagement and real deployments. The science ministry and KISA opened the country's first transition competition and hackathon on 1 October [32]. It runs until 30 October, with awards on 26 November [32]. The national patent office plans to use the Korean algorithms on an analysis platform launching in February 2027 [33].

Europe's news was mostly about coordination. On 3 September the G7 Cybersecurity Working Group, with France's ANSSI among its authors, called for immediate action on the transition [34]. ETSI's quantum-safe group has new draft parts of its TR 104 239 implementation guidance, covering ML-KEM, ML-DSA and SLH-DSA [35]. As of early September their status ranged from early draft to publication [35].

6. Isogenies, Lattice Variants and Symmetric Research

The remaining results test less visible corners of the field: subverted devices, unusual lattices, and the symmetric building blocks every scheme depends on.

The most unusual paper of the month targets CSIDH, an isogeny-based key exchange. Kleptography is the craft of hiding a backdoor that the victim cannot detect from the outputs alone. A University of Manchester researcher shows how a subverted device can leak a shared CSIDH curve through its owner's public curves in later sessions [36]. The work adapts the classic kleptographic attack on Diffie-Hellman to CSIDH [36]. It proves the subverted curves are computationally indistinguishable from honest ones, relying only on CSIDH's own assumptions [36].

Other isogeny work sized and mapped the field. A quantum security analysis of unrestricted isogeny group actions suggests 2048-bit fields for qt-Pegasis at reasonable security, and 4096-bit fields for NIST Level 1 [37]. A 10 September preprint improves supersingularity testing for twisted Hessian curves [38]. It shows that for primes congruent to 2 modulo 3, every ordinary such curve lies on the surface of the 3-isogeny volcano [38].

A lattice-isomorphism design lost much of its margin. A preprint titled "Poisoned Cryptography" attacks a lattice isomorphism scheme built from a direct sum of two rescaled lattices, solving it in half the dimension [39]. At dimension 1024, it estimates that key recovery falls from 220 to 124 classical bits [39].

On the symmetric side, two papers studied sums of random permutations against quantum attackers. One proves that the XOR of two or more independent random permutations looks like a random function to any quantum algorithm whose number of queries is at most $2^{n/57774}$ [40]. Its authors call it the first construction from permutations with quantum security beyond the birthday bound [40]. The other gives a Fourier-analytic treatment of the same question over a finite abelian group [41]. A third paper proves the two-round Even-Mansour cipher secure against superposition queries, and shows any single-permutation construction is not [42].

Hash-based signatures gained both a construction and a limit. PRAWNS is a threshold framework whose signatures look exactly like ordinary ones and reveal nothing about the parties or the threshold [43]. A separate result proves that, beyond a certain number of signers, efficient threshold signing of XMSS or SLH-DSA cannot treat the hash function as a black box [44]. On 30 September a forum post proposed executable conformance tests for stateful hash-based key management, as input to NIST's revision of SP 800-208 [45]. It notes that RFC 10033 says what a deployment must guarantee about key state, but gives no way to check it [45].

7. Where Things Stood Before This Update

This is the earlier picture, as of the last full edition at the end of August, for anyone who has not read it.

Before September, the field rested on three published standards. NIST issued FIPS 203, 204 and 205 in August 2024, for ML-KEM, ML-DSA and SLH-DSA [46]. ML-KEM and ML-DSA both rest on structured lattices, so much of the field's work was a hedge against that single assumption. NIST picked HQC in March 2025 as a code-based backup with different mathematics [47]. Its draft standard had been expected around March 2026 [47].

NIST's second signature competition existed to diversify away from lattices. In May it advanced nine candidates to round 3, only one of them lattice-based [48]. That one, HAWK, was withdrawn in July after an AI-assisted attack halved its security margin [19]. Four of the remaining candidates were multivariate, and attacks on the UOV family had widened through the year [49]. An August extension cut SNOVA's lowest level to 2^{103} gate operations, below its target [49].

Classic McEliece was the conservative holdout, standardized by ISO after NIST declined it [4]. Two August papers narrowed its margin without breaking it [50] [51]. One costed key-recovery attacks below NIST's classical reference levels, at $2^{126.77}$ bit operations for the smallest set [50]. The other gave a distinguisher running in quasipolynomial time, in an asymptotic regime [51].

Outside the United States, divergence was the theme. South Korea picked its own four winners in January 2025: HAETAE and AIMer for signatures, SMAUG-T and NTRU+ for key encapsulation [52]. China expected its own standards within three years and had selected nothing from its 2025 call [53]. Germany's guideline backed ML-KEM and FrodoKEM, and required the most sensitive uses to migrate by the end of 2030 [3]. Sole use of classical key agreement ends after 2031 [3].

France went further on enforcement. Its cybersecurity agency said it would stop certifying products without quantum-resistant encryption from 2027 [54]. FrodoKEM, the unstructured lattice alternative that both Germany and France favour, was standardized by ISO in June 2026 [55]. Japan added ML-KEM to its recommended ciphers list in March 2026 [56].

Symmetric cryptography was the settled corner. Grover's algorithm halves the effective strength of a key, so AES-256 keeps about 128 bits against a quantum attacker [57]. The NSA requires 256-bit AES keys at every classification level and treats that margin as sufficient for the foreseeable future [57]. That was the baseline this month's attacks and decisions moved against.

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